

5. Vectors

What students need to learn:

The vector product $\mathbf{a} \times \mathbf{b}$ and the triple scalar product $\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}$.

Use of vectors in problems involving points, lines and planes.

The equation of a line in the form $(\mathbf{r} - \mathbf{a}) \times \mathbf{b} = \mathbf{0}$.

The equation of a plane in the forms $\mathbf{r} \cdot \mathbf{n} = p$, $\mathbf{r} = \mathbf{a} + s\mathbf{b} + t\mathbf{c}$.

The interpretation of $|\mathbf{a} \times \mathbf{b}|$ as an area and $\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}$ as a volume.

Students may be required to use equivalent cartesian forms also.

Applications to include

- (i) distance from a point to a plane,
- (ii) line of intersection of two planes,
- (iii) shortest distance between two skew lines.

Students may be required to use equivalent cartesian forms also.
